

The topology of software risk in scientific models

5. Expert questionnaire validation

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1 Preliminary

```
# PRELIMINARY FUNCTIONS #####
#####

sensobol::load_packages(c("data.table", "readxl", "ggplot2", "cowplot",
                          "softwareRisk", "parallel", "benchmarkme"))

theme_AP <- softwareRisk::theme_AP

# Load data -----

tidy <- as.data.table(read_excel("./questionnaire_data/questionnaire_analysis.xlsx",
                                sheet = "1_TidyData"))

setnames(tidy,
  old = seq_len(14),
  new = c("model", "respondent_id", "path_num", "path_name",
          "path_type", "familiarity_text", "familiarity_num",
          "q1_real_execution", "q2_central_workflow",
          "q3_maintenance", "q4_system_consequences",
          "mean_likert", "excluded_unfamiliar", "comment"))

usable <- tidy[excluded_unfamiliar == FALSE]

usable[, path_type := factor(path_type, levels = c("high_risk", "low_risk"),
                             labels = c("High-risk", "Low-risk"))]
usable[, model := factor(model, levels = c("CTSM", "H08", "HydroPy", "HYPE", "PCRGLOBWB"),
                          labels = c("CTSM", "H08", "HydroPy", "HYPE", "PCR-GLOBWB"))]

q_cols <- c("q1_real_execution", "q2_central_workflow",
            "q3_maintenance", "q4_system_consequences")
q_names <- c("Q1: Real execution chain",
             "Q2: Central to workflow",
             "Q3: Maintenance relevance",
             "Q4: System-wide consequences")
```

2 Sample summary

```
# SAMPLE #####

cat("-----\n")

## -----

cat("SAMPLE\n")

## SAMPLE
```

```

cat("-----\n")

## -----

cat("Total records (all) :", nrow(tidy), "\n")

## Total records (all) : 107

cat("Excluded (unfamiliar):", nrow(tidy[excluded_unfamiliar == TRUE]), "\n")

## Excluded (unfamiliar): 18

cat("Usable records :", nrow(usable), "\n\n")

## Usable records : 87

print(usable[, .N, by = .(model, path_type)])

##           model path_type      N
##           <fctr>   <fctr> <int>
## 1:         CTSM High-risk     23
## 2:         CTSM  Low-risk      4
## 3:          H08 High-risk     10
## 4:          H08  Low-risk      5
## 5:      HydroPy High-risk     10
## 6:      HydroPy  Low-risk      5
## 7:          HYPE High-risk     10
## 8:          HYPE  Low-risk      5
## 9: PCR-GLOBWB High-risk     10
## 10: PCR-GLOBWB  Low-risk      5

```

3 Descriptive statistics

```

# DESCRIPTIVE STATISTICS #####

cat("-----\n")

## -----

cat("DESCRIPTIVES: mean_likert (all models pooled)\n")

## DESCRIPTIVES: mean_likert (all models pooled)

cat("-----\n")

## -----

desc <- usable[, .(
  N      = sum(!is.na(mean_likert)),
  Mean   = round(mean(mean_likert, na.rm = TRUE), 3),
  Median = round(median(mean_likert, na.rm = TRUE), 3),
  SD     = round(sd(mean_likert, na.rm = TRUE), 3),

```

```

    Q25    = round(quantile(mean_likert, .25, na.rm = TRUE), 3),
    Q75    = round(quantile(mean_likert, .75, na.rm = TRUE), 3)
  ), by = path_type]
print(desc)

```

```

##   path_type      N Mean Median    SD  Q25  Q75
##   <fctr> <int> <num> <num> <num> <num> <num>
## 1: High-risk   63 3.187    3.5 1.063  2.5 4.000
## 2: Low-risk    23 3.304    3.5 0.482  3.0 3.625

```

```
cat("\n")
```

```
cat("Per-question means by path type:\n")
```

```
## Per-question means by path type:
```

```

for (i in seq_along(q_cols)) {
  hr <- usable[path_type == "High-risk", get(q_cols[i])]
  lr <- usable[path_type == "Low-risk",  get(q_cols[i])]
  cat(sprintf("  %-35s High = %.3f Low = %.3f Diff = %+.3f\n",
              q_names[i], mean(hr, na.rm=T), mean(lr, na.rm=T),
              mean(hr, na.rm=T) - mean(lr, na.rm=T)))
}

```

```

##   Q1: Real execution chain          High = 3.460 Low = 3.783 Diff = -0.322
##   Q2: Central to workflow          High = 3.556 Low = 3.391 Diff = +0.164
##   Q3: Maintenance relevance       High = 2.413 Low = 2.522 Diff = -0.109
##   Q4: System-wide consequences     High = 3.317 Low = 3.522 Diff = -0.204

```

4 Primary test

```
# PRIMARY TEST #####
```

```
cat("-----\n")
```

```
## -----
```

```
cat("PRIMARY: Wilcoxon rank-sum, mean_likert, all models pooled\n")
```

```
## PRIMARY: Wilcoxon rank-sum, mean_likert, all models pooled
```

```
cat("H1: high-risk > low-risk (one-sided)\n")
```

```
## H1: high-risk > low-risk (one-sided)
```

```
cat("-----\n")
```

```
## -----
```

```
hr_all <- usable[path_type == "High-risk", mean_likert]
```

```
lr_all <- usable[path_type == "Low-risk", mean_likert]
```

```

wt_all <- wilcox.test(hr_all, lr_all, alternative = "greater", exact = FALSE)
A_all <- wt_all$statistic / (length(hr_all) * length(lr_all))

cat(sprintf(" W = %.0f\n p = %.4f (one-sided, high > low)\n A = %.3f\n",
            wt_all$statistic, wt_all$p.value, A_all))

## W = 736
## p = 0.4589 (one-sided, high > low)
## A = 0.486

cat(sprintf(" n_high = %d, n_low = %d\n", length(hr_all), length(lr_all)))

## n_high = 63, n_low = 24

```

5 Per-question tests

```

# PER-QUESTION TESTS #####

cat("-----\n")

## -----

cat("PER-QUESTION Wilcoxon rank-sum (one-sided), all models\n")

## PER-QUESTION Wilcoxon rank-sum (one-sided), all models

cat("-----\n")

## -----

pq_results <- lapply(seq_along(q_cols), function(i) {
  hr <- usable[path_type == "High-risk", get(q_cols[i])]
  lr <- usable[path_type == "Low-risk", get(q_cols[i])]
  wt <- wilcox.test(hr, lr, alternative = "greater", exact = FALSE)
  A <- wt$statistic / (sum(!is.na(hr)) * sum(!is.na(lr)))
  list(q = q_names[i], W = wt$statistic, p = wt$p.value, A = A,
       mean_hr = mean(hr, na.rm=T), mean_lr = mean(lr, na.rm=T))
})

raw_p <- sapply(pq_results, `[`, "p")
adj_p <- p.adjust(raw_p, method = "BH")

for (i in seq_along(pq_results)) {
  r <- pq_results[[i]]
  cat(sprintf(" %-35s W=%.0f p=%.4f p_BH=%.4f A=%.3f Diff=%+.3f\n",
              r$q, r$W, r$p, adj_p[i], r$A, r$mean_hr - r$mean_lr))
}

```

```

## Q1: Real execution chain W= 677 p=0.6883 p_BH=0.7351 A=0.467 Diff=-0.322
## Q2: Central to workflow W= 811 p=0.1929 p_BH=0.7351 A=0.560 Diff=+0.164

```

```
## Q3: Maintenance relevance W= 664 p=0.7351 p_BH=0.7351 A=0.458 Diff=-0.109
## Q4: System-wide consequences W= 726 p=0.4980 p_BH=0.7351 A=0.501 Diff=-0.204
```

6 Per-model tests

```
# PER-MODEL TESTS #####
```

```
cat("-----\n")
```

```
## -----
```

```
cat("PER-MODEL Wilcoxon rank-sum, mean_likert (two-sided)\n")
```

```
## PER-MODEL Wilcoxon rank-sum, mean_likert (two-sided)
```

```
cat("-----\n")
```

```
## -----
```

```
for (m in levels(usable$model)) {
  sub_m <- usable[model == m]
  hr_m <- sub_m[path_type == "High-risk", mean_likert]
  lr_m <- sub_m[path_type == "Low-risk", mean_likert]
  wt_m <- wilcox.test(hr_m, lr_m, alternative = "two.sided", exact = FALSE)
  A_m <- wt_m$statistic / (length(hr_m) * length(lr_m))
  cat(sprintf(" %-12s W=%5.0f p=%.4f A=%.3f mean_high=%.2f mean_low=%.2f n_high=%d n_low=%d\n",
              m, wt_m$statistic, wt_m$p.value, A_m,
              mean(hr_m, na.rm=T), mean(lr_m, na.rm=T),
              length(hr_m), length(lr_m)))
}
```

```
## CTSM W= 54 p=0.6239 A=0.582 mean_high=3.47 mean_low=3.38 n_high=23 n_low=23
## H08 W= 38 p=0.1204 A=0.760 mean_high=4.00 mean_low=3.55 n_high=10 n_low=10
## HydroPy W= 28 p=0.7543 A=0.560 mean_high=3.08 mean_low=2.95 n_high=10 n_low=10
## HYPE W= 0 p=0.0029 A=0.010 mean_high=1.85 mean_low=3.35 n_high=10 n_low=10
## PCR-GLOBWB W= 24 p=0.6143 A=0.480 mean_high=3.17 mean_low=3.31 n_high=10 n_low=10
```

7 Sensitivity: exclude HYPE

```
# SENSITIVITY: EXCLUDE HYPE #####
```

```
cat("-----\n")
```

```
## -----
```

```
cat("SENSITIVITY: Exclude HYPE (deprecated calibration paths)\n")
```

```
## SENSITIVITY: Exclude HYPE (deprecated calibration paths)
```

```
cat("-----\n")
```

```
## -----
no_hype <- usable[model != "HYPE"]
hr_nh <- no_hype[path_type == "High-risk", mean_likert]
lr_nh <- no_hype[path_type == "Low-risk", mean_likert]
wt_nh <- wilcox.test(hr_nh, lr_nh, alternative = "greater", exact = FALSE)
A_nh <- wt_nh$statistic / (length(hr_nh) * length(lr_nh))

cat(sprintf(" W = %.0f\n p = %.4f (one-sided)\n A = %.3f\n",
            wt_nh$statistic, wt_nh$p.value, A_nh))

## W = 588
## p = 0.0701 (one-sided)
## A = 0.584

cat(sprintf(" n_high = %d, n_low = %d\n", length(hr_nh), length(lr_nh)))

## n_high = 53, n_low = 19

cat(sprintf(" mean_high = %.3f, mean_low = %.3f\n\n",
            mean(hr_nh, na.rm=T), mean(lr_nh, na.rm=T)))

## mean_high = 3.439, mean_low = 3.292

cat("Per-question (excluding HYPE):\n")

## Per-question (excluding HYPE):
for (i in seq_along(q_cols)) {
  hr <- no_hype[path_type == "High-risk", get(q_cols[i])]
  lr <- no_hype[path_type == "Low-risk", get(q_cols[i])]
  wt <- wilcox.test(hr, lr, alternative = "greater", exact = FALSE)
  A <- wt$statistic / (sum(!is.na(hr)) * sum(!is.na(lr)))
  cat(sprintf(" %-35s p=%.4f A=%.3f Diff=%+.3f\n",
              q_names[i], wt$p.value, A, mean(hr, na.rm=T)-mean(lr, na.rm=T)))
}

## Q1: Real execution chain p=0.1669 A=0.572 Diff=+0.089
## Q2: Central to workflow p=0.1647 A=0.575 Diff=+0.233
## Q3: Maintenance relevance p=0.3615 A=0.527 Diff=+0.086
## Q4: System-wide consequences p=0.1193 A=0.589 Diff=+0.180
```

8 Familiarity as moderator

```
# FAMILIARITY CORRELATION #####
cat("-----\n")

## -----
cat("FAMILIARITY CORRELATION with mean_likert\n")
```

```
## FAMILIARITY CORRELATION with mean_likert
cat("-----\n")

## -----
fam_cor <- cor.test(usable$familiarity_num, usable$mean_likert,
                    method = "spearman", exact = FALSE)
cat(sprintf(" Spearman rho = %.3f, p = %.4f\n",
            fam_cor$estimate, fam_cor$p.value))

## Spearman rho = -0.026, p = 0.8150
```

9 Plots

```
# PLOT DISTRIBUTIONS #####
```

```
p1 <- ggplot(usable, aes(x = path_type, y = mean_likert,
                        fill = path_type, colour = path_type)) +
  geom_violin(alpha = 0.35, trim = FALSE, width = 0.8) +
  geom_boxplot(width = 0.15, outlier.shape = NA, alpha = 0.8,
              colour = "grey30") +
  geom_jitter(width = 0.08, size = 1.5, alpha = 0.6) +
  scale_fill_manual(values = c("High-risk" = "#2471A3",
                              "Low-risk" = "#E67E22")) +
  scale_colour_manual(values = c("High-risk" = "#2471A3",
                                 "Low-risk" = "#E67E22")) +
  scale_y_continuous(limits = c(0.5, 5.5), breaks = 1:5) +
  labs(x = NULL, y = "Mean Likert score",
       title = "All models",
       subtitle = sprintf("Wilcoxon W = %.0f, p = %.3f, \nA = %.3f",
                          wt_all$statistic, wt_all$p.value, A_all)) +
  theme_AP() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold"),
        plot.subtitle = element_text(size = 6, colour = "grey40"))

p1
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_ydensity()`).
```

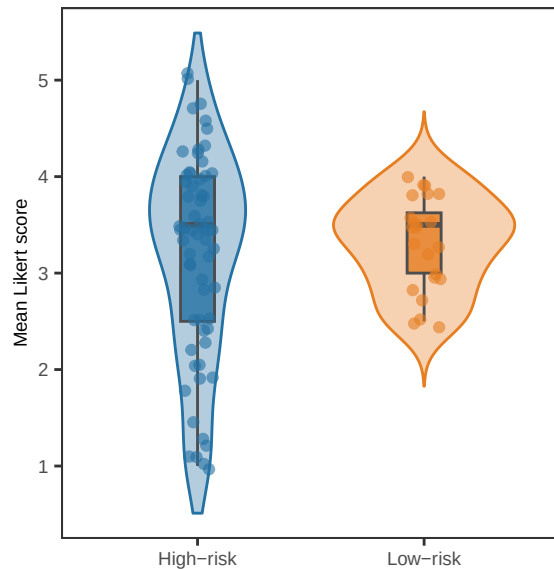
```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 120 rows containing missing values or values outside the scale range
## (`geom_violin()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```


All models

Wilcoxon W = 736, p = 0.459,
A = 0.486



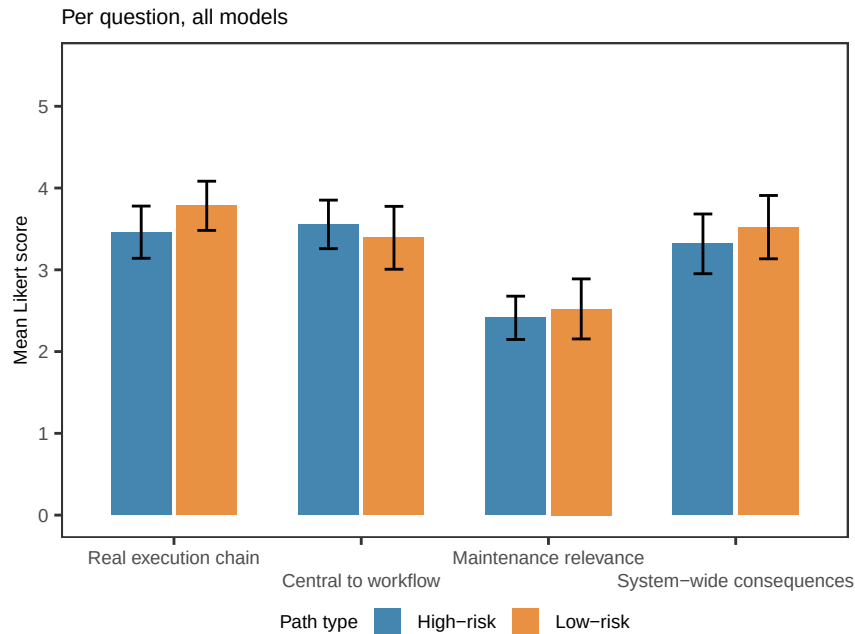
PLOT PER QUESTION

```
q_long <- melt(usable, measure.vars = q_cols,
               variable.name = "question", value.name = "score")
q_long[, question := factor(question, levels = q_cols, labels = q_names)]
q_long <- q_long[!is.na(score)]

q_desc <- q_long[, .(mean = mean(score),
                    lower = mean(score) - 1.96 * sd(score) / sqrt(.N),
                    upper = mean(score) + 1.96 * sd(score) / sqrt(.N)),
                  by = .(question, path_type)]

p3 <- ggplot(q_desc, aes(x = question, y = mean, fill = path_type,
                        ymin = lower, ymax = upper)) +
  geom_col(position = position_dodge(0.7), width = 0.65, alpha = 0.85) +
  geom_errorbar(position = position_dodge(0.7), width = 0.2,
               linewidth = 0.5) +
  scale_fill_manual(values = c("High-risk" = "#2471A3",
                              "Low-risk" = "#E67E22"),
                   name = "Path type") +
  scale_y_continuous(limits = c(0, 5.5), breaks = 0:5) +
  scale_x_discrete(labels = function(x) sub("Q\\d+": "", x),
                  guide = guide_axis(n.dodge=2)) +
  labs(x = NULL, y = "Mean Likert score",
       title = "Per question, all models") +
  theme_AP() +
  theme(legend.position = "bottom")
```

p3



PLOT PER MODEL

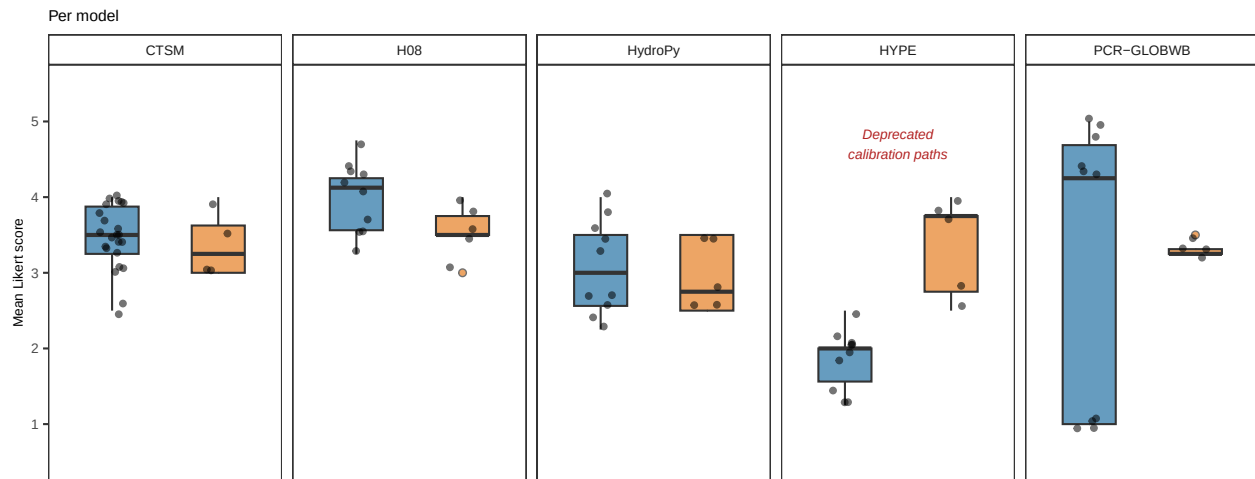
```
p4 <- ggplot(usable,
  aes(x = path_type, y = mean_likert, fill = path_type)) +
  geom_boxplot(width = 0.5, outlier.shape = 21, outlier.size = 1.5,
    alpha = 0.7) +
  geom_jitter(width = 0.12, size = 1.2, alpha = 0.55) +
  scale_fill_manual(values = c("High-risk" = "#2471A3",
    "Low-risk" = "#E67E22"),
    name = "Path type") +
  scale_y_continuous(limits = c(0.5, 5.5), breaks = 1:5) +
  facet_wrap(~model, nrow = 1) +
  geom_text(data = data.frame(model = factor("HYPE", levels = levels(usable$model)),
    x = 1.5, y = 4.7),
    aes(x = x, y = y, label = "Deprecated\ncalibration paths"),
    inherit.aes = FALSE,
    size = 2.5, colour = "firebrick", fontface = "italic") +
  labs(x = NULL, y = "Mean Likert score",
    title = "Per model") +
  theme_AP() +
  theme(axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    legend.position = "none")

p4
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```



```
# ASSEMBLE FIGURE #####
```

```
top <- plot_grid(p1, p3, nrow = 1, rel_widths = c(0.3, 0.7),
  labels = c("a", "b"))
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_ydensity()`).
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 120 rows containing missing values or values outside the scale range
## (`geom_violin()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
full <- plot_grid(top, p4, ncol = 1, rel_heights = c(0.6, 0.4),
  labels = c("", "c"))
```

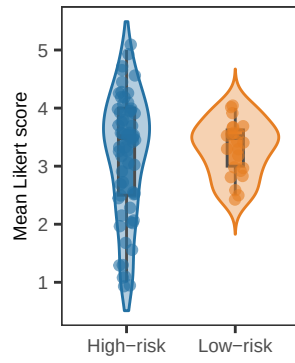
```
## Warning: Removed 1 row containing non-finite outside the scale range (`stat_boxplot()`).
## Removed 1 row containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```

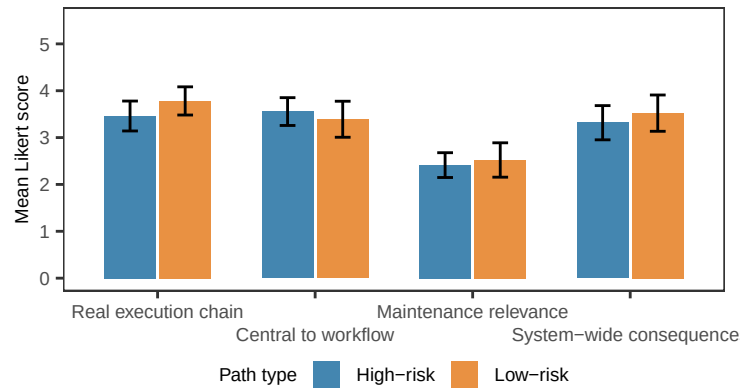
```
full
```

a All models

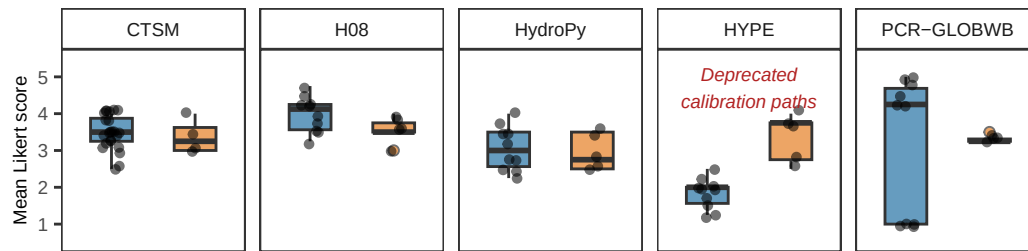
Wilcoxon W = 736, p = 0.459,
A = 0.486



b Per question, all models



c Per model



10 Session information

```
# SESSION INFORMATION #####
```

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] parallel stats graphics grDevices utils datasets methods
## [8] base
##
## other attached packages:
## [1] benchmarkme_1.0.8 softwareRisk_0.2.0 cowplot_1.2.0.9000
## [4] ggplot2_4.0.1.9000 readxl_1.4.5 data.table_1.18.2.1
##
## loaded via a namespace (and not attached):
## [1] benchmarkmeData_1.0.4 gtable_0.3.6 xfun_0.55
## [4] ggrepel_0.9.6 lattice_0.22-7 vctrs_0.7.1
## [7] tools_4.5.2 Rdpack_2.6.4 generics_0.1.4
## [10] tibble_3.3.1 pkgconfig_2.0.3 Matrix_1.7-4
## [13] RColorBrewer_1.1-3 S7_0.2.1 lifecycle_1.0.5
## [16] ineq_0.2-13 compiler_4.5.2 farver_2.1.2
## [19] tinytex_0.58 ggforce_0.5.0 graphlayouts_1.2.2
## [22] codetools_0.2-20 htmltools_0.5.9 yaml_2.3.12
## [25] pillar_1.11.1 tidyr_1.3.2 MASS_7.3-65
## [28] cachem_1.1.0 iterators_1.0.14 viridis_0.6.5
## [31] foreach_1.5.2 tidyselect_1.2.1 digest_0.6.39
## [34] dplyr_1.1.4 purrr_1.2.1 labeling_0.4.3
## [37] polyclip_1.10-7 sensobol_1.1.6 fastmap_1.2.0
## [40] grid_4.5.2 cli_3.6.5 magrittr_2.0.4
## [43] ggraph_2.2.2 tidygraph_1.3.1 withr_3.0.2
## [46] scales_1.4.0 rmarkdown_2.30 httr_1.4.8
## [49] igraph_2.2.1 otel_0.2.0 gridExtra_2.3
## [52] cellranger_1.1.0 memoise_2.0.1 evaluate_1.0.5
## [55] knitr_1.51 rbibutils_2.4 doParallel_1.0.17
```

```
## [58] viridisLite_0.4.2      rlang_1.1.7             Rcpp_1.1.1
## [61] glue_1.8.0              tweenr_2.0.3            rstudioapi_0.17.1
## [64] R6_2.6.1
```

```
## Return the machine CPU -----
```

```
cat("Machine:    "); print(get_cpu()$model_name)
```

```
## Machine:
```

```
## [1] "Apple M1 Max"
```

```
## Return number of true cores -----
```

```
cat("Num cores:   "); print(parallel::detectCores(logical = FALSE))
```

```
## Num cores:
```

```
## [1] 10
```

```
## Return number of threads -----
```

```
cat("Num threads: "); print(parallel::detectCores(logical = FALSE))
```

```
## Num threads:
```

```
## [1] 10
```